Homework 2

BME 5910

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Optical Clearing

Optical clearing is a technique that allows us to make a sample more transparent to the optical system that is viewing into the material. Fundamentally, what makes an object difficult to image is how well-behaved light is inside, including scattering and absorption. If too much light is absorbed within the material, then the size of the exiting signal is very low. Similarly, if too much light is scattered, then rays that initially are focusing to a point within the medium don't retain as much information as they do before. Optical clearing reduces scattering in order to improve image quality and allow for deeper penetration into the tissue.

A photon scatters because it reaches a boundary between different refractive indices. Based on Fresnel's equations, we know that at this boundary, some light is reflected and some is refracted. Unless the light hits this surface obliquely (perpendicular to the surface), the direction of the photon's path will change. Additionally, the difference in refractive index, n, between the two mediums determines the strength of the reflection and refraction angle. In tissue, like many mediums, has different components that create many boundaries within the medium. These boundaries are at random angles, which means that chances are that the light will significantly change direction after few scattering events. For example, skin is made of water, fat, proteins, and blood vessels. When light enters the skin, it mainly travels through water and then scatters when it hits a fat deposit for example.

The main way optical clearing works is by reducing the mismatch in refractive index. Numerically, we can derive the scattering efficiency, Q_s, for a spherical particle in the Rayleigh regime.

$$Q_s = \frac{8x^4}{3}\left(\frac{n_{\text{rel}}^2 - 1}{n_{\text{rel}}^2 + 2}\right)^2 \text{ where } n_{\text{rel}} = \frac{n_{\text{scatterer}}}{n_{\text{background}}}.$$

Here, we see that as n_{rel} gets closer to 1, Q_s limits to zero. So, by increasing the background refractive index (typically water around 1.33) closer to that of fat droplets (~1.47), we can reduce the overall scattering coefficient of the tissue.

To implement this, an optical clearing agent (OCA) is absorbed into the tissue before imaging. This fills the water with a concentration of fluid with a higher index of refraction and thus reduces scattering in the tissue. This makes the diffusion-dominated region start deeper within the tissue and gives more useful images. This can be applied to both surface and volume images. For example, a topical clearing agent can be applied to the skin of a mouse for example and that can increase the depth that optical coherence tomography (OCT) or other optical images can penetrate the surface tissue. Another application can be for imaging entire organs. Here, the organ to be studied can be soaked in the OCA for days-months before imaging.

Rayleigh Calculation

```
% Inputs (SI units)
d = 20e-9; % [m] Diameter of spherical scatterer
a = d / 2; % [m] Radius of spherical scatterer
lambda = 400e-9; % [m] Wavelength of light
n_sphere = 1.57; % Refractive index of the spherical scatterer
n_background = 1.33; % Refractive index of the background
w_sphere = 1.05e3; % [g/cm3] Density of sphere
w_background = 1e3; % [g/cm3] Density of background
C_wt = 1e-5; % [mass sphere / mass background] Concentration of sphere in
background by weight (mass)

% Calculations
k=2*pi*n_background/lambda % Wave number of light in background
```

```
k =
2.0892e+07
```

```
x=k*a % Optical size parameter (should be << 1 for Rayleigh)
```

```
x =
0.2089
```

```
n_rel=n_sphere/n_background % Ratio of refractive indices
```

```
n_rel =
1.1805
```

```
V_s = 4/3*pi*a^3 % [m3] Volume of a sphere
```

```
V_s =
4.1888e-24
```

```
sigma_s = 8*pi*a^2*x^4/3*((n_rel^2 - 1)/(n_rel^2+2))^2 % [m2] Scattering cross
section
```

```
sigma_s =
2.1455e-20
```

```
Q_s = sigma_s/pi/a^2 % [-] Scattering efficiency (of a single scatterer)
```

```
Q_s =
6.8294e-05
```

```
N_s = C_wt*(w_background/w_sphere)/V_s % [1/m3] Number density of scatterers
```

```
N_s =
2.2736e+18
```

```
mu_s = N_s * sigma_s % [1/m] Scattering coefficient (Beer's law)
```

```
mu_s =
0.0488
```

```
% Plot
x_range = logspace(-1,0,50);
Q_s_range = 8*x_range.^4/3*((n_rel^2 - 1)/(n_rel^2+2))^2;
plot(x_range,Q_s_range, "--","DisplayName","Rayleigh Approximation")
hold on;
plot(x,Q_s,".r", "MarkerSize",10, "DisplayName","Example Point")
hold off;
yscale("log")
xscale("log")
xlabel("Optical size parameter, x = ka")
ylabel("Scattering efficiency, Q_s")
title("Rayleigh Scattering Regime")
legend("Location","northwest")
```

